

The Neutrino Packages

dist · poly · finance · astro · rmt · phys · scatter · symb · demo

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Contents

A package is a file of let definitions; `load("packages/name.nu")` runs it in the current session and its bindings persist. Records of closures act as namespaces (`norm.cdf`), and packages validate their inputs with `assert`, so they fail like builtins do — with a message, not an accident. Every transcript below is machine-verified against the interpreter by `tests/run_manual.sh`, and every numerical claim is cross-checked in the package's own golden tests against an independent reference (SciPy, NumPy, Python's `datetime`, or the `astral` library — named in each section).

1. dist.nu — probability distributions

Six distributions — `norm`, `student`, `chi2`, `fdist`, `expo`, `unif` — each a record with `pdf`, `cdf`, `inv` (quantile), and `rand`. Parameters are explicit: `norm.cdf(x, mu, sigma)`. Quantiles with no closed form use bracket expansion plus `fzero` on the CDF; the quantile domain is $0 < p < 1$. Cross-checked against SciPy (`tests/34_dist.test`).

```
neutrino> load("packages/dist.nu")
neutrino> norm.cdf(1.96, 0, 1)
0.975002
neutrino> norm.inv(0.975, 0, 1)
1.95996
neutrino> student.inv(0.975, 30)
2.04227
neutrino> chi2.inv(0.95, 3)
7.81473
neutrino> fdist.cdf(3.0, 4, 20)
0.956799
```

A complete two-sample t-test — simulate, test, decide:

```
neutrino> load("packages/dist.nu")
neutrino> rng(42)
neutrino> let x1 = norm.rand(40, 100, 15)
[75.8016; 123.017; 111.725; 93.9971; 100.238; 98.0904; 107.158; 90.1486; 90.4082; 94.4609; 96.6851;
neutrino> let x2 = norm.rand(40, 108, 15)
[102.897; 92.0654; 112.122; 98.9768; 112.891; 112.931; 92.8931; 102.642; 111.903; 100.646; 100.94;
neutrino> let t = (mean(x2) - mean(x1)) / sqrt(var(x1)/40 + var(x2)/40)
2.46819
neutrino> 2 * (1 - student.cdf(abs(t), 78))
```

0.0157683

Out-of-domain probabilities refuse with the package's own message:

```
neutrino> load("packages/dist.nu")
error: quantile: p must be in (0,1), got 1.5
neutrino> student.inv(1.5, 10)
```

Function	Worked example	Result
norm.pdf/cdf/inv(x, mu, sigma)	norm.cdf(1.96, 0, 1)	0.975002
norm.rand(n, mu, sigma)	mean(norm.rand(500, 10, 2)) > 9	true
student.pdf/cdf/inv(x, v)	student.inv(0.975, 30)	2.04227
chi2.pdf/cdf/inv(x, k)	chi2.inv(0.95, 3)	7.81473
fdist.pdf/cdf/inv(x, d1, d2)	fdist.cdf(3.0, 4, 20)	0.956799
expo.pdf/cdf/inv(x, rate)	expo.inv(0.5, 2)	0.346574
unif.pdf/cdf/inv(x, a, b)	unif.cdf(3, 0, 10)	0.300000

2. poly.nu — polynomials

Coefficients are row vectors, highest power first: [2, -3, 1] is $2x^2 - 3x + 1$. Roots come from the companion matrix and eig — the same algorithm Octave uses, on Neutrino's LAPACK-verified eigensolver. Cross-checked against NumPy (tests/37_poly.test).

```
neutrino> load("packages/poly.nu")
neutrino> polyval([2, -3, 1], 4)
21
neutrino> roots([1, -6, 11, -6])'
[1, 2, 3]
neutrino> roots([1, 0, 1])'
[1i, -1i]
neutrino> companion([1, -6, 11, -6])
[6, -11, 6; 1, 0, 0; 0, 1, 0]
```

Least-squares fitting (Vandermonde + backslash), and calculus that inverts:

```
neutrino> load("packages/poly.nu")
neutrino> let x = [0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
neutrino> let y = [1.1, 2.9, 9.2, 19.1, 32.8]
[1.1, 2.9, 9.2, 19.1, 32.8]
neutrino> polyfit(x, y, 2)
[1.95714, 0.131429, 1.01429]
neutrino> polyder([1, -6, 11, -6])
[3, -12, 11]
neutrino> polyint(polyder([1, -6, 11, -6]), -6)
[1, -6, 11, -6]
neutrino> conv([1, -1], [1, -2])
[1, -3, 2]
```

Function	Worked example	Result
polyval(c, x)	polyval([2, -3, 1], 4)	21
roots(c)	sort(real(roots([1, -6, 11, -6])))'	[1.00000, 2.00000, 3.00000]
companion(c)	companion([1, 0, -4])	[0.00000, 4.00000; 1.00000, 0.00000]
polyfit(x, y, n)	polyfit([1, 2, 3], [2, 5, 10], 2)	[1.00000, -8.32667e-16, 1.00000]
polyder(c)	polyder([1, -6, 11, -6])	[3, -12, 11]
polyint(c, k)	polyint([3, -12, 11], -6)	[1.00000, -6.00000, 11.0000, -6.00000]
conv(a, b)	conv([1, -1], [1, -2])	[1.00000, -3.00000, 2.00000]

3. finance.nu — the HP-12C's greatest hits

Time value of money, cash flows, bonds, amortization, and date arithmetic. Sign convention (HP-12C): money received is positive, money paid is negative; rates are per period, as decimals. Cross-checked against SciPy and Python's datetime (tests/38_finance.test).

The mortgage block — payment, implied rate, savings growth:

```
neutrino> load("packages/finance.nu")
neutrino> pmt(360, 0.005, 250000, 0)
-1498.88
neutrino> rate(360, 250000, -1498.876313, 0)
0.005
neutrino> fv(120, 0.004, 0, -200)
30726.4
```

Cash flows and bonds:

```
neutrino> load("packages/finance.nu")
neutrino> npv(0.10, [-1000, 300, 420, 680])
130.729
neutrino> irr([-1000, 300, 420, 680])
0.163406
neutrino> bond_price(100, 0.08, 0.06, 10, 1)
114.72
neutrino> bond_ytm(114.7202, 100, 0.08, 10, 1)
0.06
neutrino> bond_mduration(100, 0.08, 0.06, 10, 1)
7.0236
```

amort returns the whole schedule as a record of columns — a small data frame you can index, sum, and plot:

```
neutrino> load("packages/finance.nu")
neutrino> let s = amort(1000, 0.01, 3)
{period = [1; 2; 3], payment = [340.022; 340.022; 340.022], interest = [10; 6.69978; 3.36656], principal = [12]}
neutrino> fields(s)'
["period", "payment", "interest", "principal", "balance"]
neutrino> s.payment[1]
```

```
340.022
neutrino> s.balance'
[669.978, 336.656, 4.26326e-12]
neutrino> sum(s.principal)
1000
```

Dates, HP-12C style — actual day counts, date arithmetic, day of week, and the bond market's 30/360 convention. `datestr` returns a display string, `daterec` the {y, m, d} record, and `today()` a serial day number, so arithmetic reads naturally — `datestr(today() + 90)` is the date in 90 days:

```
neutrino> load("packages/finance.nu")
neutrino> days(2026, 1, 1, 2026, 7, 9)
189
neutrino> datestr(datenum(2026, 7, 17) + 90)
"2026-10-15"
neutrino> daterec(datenum(2024, 2, 29))
{y = 2024, m = 2, d = 29}
neutrino> dateadd(2024, 2, 28, 2)
{y = 2024, m = 3, d = 1}
neutrino> dow(2026, 7, 9)
4
neutrino> days360(2024, 1, 31, 2024, 7, 31)
180
neutrino> daterec(today()).y >= 2026
true
```

Function	Worked example	Result
<code>pmt(n, i, pv, fv)</code>	<code>pmt(360, 0.005, 250000, 0)</code>	-1498.88
<code>pv(n, i, pmt, fv)</code>	<code>pv(360, 0.005, -1498.88, 0)</code>	250001.
<code>fv(n, i, pv, pmt)</code>	<code>fv(120, 0.004, 0, -200)</code>	30726.4
<code>nper(i, pv, pmt, fv)</code>	<code>nper(0.005, 250000, -1498.88, 0)</code>	359.998
<code>rate(n, pv, pmt, fv)</code>	<code>rate(360, 250000, -1498.876313, 0)</code>	0.00500000
<code>npv(r, cfs)</code>	<code>npv(0.10, [-1000, 300, 420, 680])</code>	130.729
<code>irr(cfs)</code>	<code>irr([-1000, 300, 420, 680])</code>	0.163406
<code>bond_price(face, crate, y, n, freq)</code>	<code>bond_price(100, 0.08, 0.06, 10, 1)</code>	114.720
<code>bond_ytm(price, face, crate, n, freq)</code>	<code>bond_ytm(114.7202, 100, 0.08, 10, 1)</code>	0.0600000
<code>bond_duration(face, crate, y, n, freq)</code>	<code>bond_duration(100, 0.08, 0.06, 10, 1)</code>	7.44502
<code>bond_mduration(face, crate, y, n, freq)</code>	<code>bond_mduration(100, 0.08, 0.06, 10, 1)</code>	7.02360
<code>bond_convexity(face, crate, y, n, freq)</code>	<code>bond_convexity(100, 0.08, 0.06, 10, 1)</code>	65.1716
<code>amort(principal, i, n)</code>	<code>amort(1000, 0.01, 3).balance[3]</code>	4.26326e-12
<code>datenum(y, m, d)</code>	<code>datenum(2026, 7, 17)</code>	2.46124e+06
<code>datestr(jdn)</code>	<code>datestr(datenum(2026, 7, 17) + 90)</code>	"2026-10-15"

Function	Worked example	Result
<code>daterec(jdn)</code>	<code>daterec(datenum(2026, 7, 17))</code>	<code>{y = 2026.00, m = 7.000000, d = 17.00000}</code>
<code>dateadd(y, m, d, k)</code>	<code>dateadd(2024, 12, 31, 1)</code>	<code>{y = 2025.00, m = 1.000000, d = 1.000000}</code>
<code>today()</code>	<code>today() == datenum(now().y, now().m, now().d)</code>	<code>true</code>
<code>dow(y, m, d)</code>	<code>dow(2026, 7, 17)</code>	<code>5.000000</code>
<code>days(y1,m1,d1, y2,m2,d2)</code>	<code>days(2026, 1, 1, 2026, 7, 17)</code>	<code>197.000</code>
<code>days360(y1,m1,d1, y2,m2,d2)</code>	<code>days360(2024, 1, 31, 2024, 7, 31)</code>	<code>180</code>

4. astro.nu — the solar almanac

Sunrise, sunset, three grades of twilight, solar noon, day length, sun position, and moon phase, for any latitude and longitude (degrees; north and east positive). Times are local decimal hours given a UTC offset `tz`; `hm` renders them as “HH:MM”. NOAA solar position algorithm, within about a minute of the astral reference library (`tests/39_astro.test`). A `places` record preloads coordinates for several cities.

```
neutrino> load("packages/astro.nu")
neutrino> hm(sunrise(places.alexandria_va.lat, places.alexandria_va.lon, 2026, 7, 10, -4))
"05:52"
neutrino> hm(sunset(places.duluth_ga.lat, places.duluth_ga.lon, 2026, 7, 10, -4))
"20:50"
neutrino> hm(solar_noon(places.ljubljana.lon, 2026, 7, 10, 2))
"13:07"
neutrino> day_length(places.ljubljana.lat, places.ljubljana.lon, 2026, 7, 10)
15.5331
```

Planning a drive to arrive before dark — civil dawn at the origin to civil dusk at the destination:

```
neutrino> load("packages/astro.nu")
neutrino> let w = drive_daylight(places.alexandria_va, places.duluth_ga, 2026, 7, 10, -4)
{depart_dawn = 5.34382, depart_rise = 5.86728, arrive_set = 20.8376, arrive_dusk = 21.3161, window_h
neutrino> hm(w.depart_dawn) + " to " + hm(w.arrive_dusk)
"05:21 to 21:19"
neutrino> w.window_hours
15.9723
```

Sun position, moon illumination, and the honest polar refusal:

```
neutrino> load("packages/astro.nu")
error: the sun does not cross 90.833 degrees at latitude 80 on 2026-6-21
neutrino> let p = sun_position(38.8048, -77.0469, 2026, 7, 10, 13.22, -4)
{alt = 73.3594, az = 179.65}
neutrino> p.alt
73.3594
neutrino> moon_illum(2026, 7, 10)
0.19449
```

```
neutrino> sunrise(80, 0, 2026, 6, 21, 0)
```

Function	Worked example	Result
sunrise(lat, lon, y, m, d, tz)	hm(sunrise(38.8048, -77.0469, 2026, 7, 17, -4))	"05:57"
sunset(lat, lon, y, m, d, tz)	hm(sunset(38.8048, -77.0469, 2026, 7, 17, -4))	"20:31"
dawn_civil / dusk_civil(lat, lon, y, m, d, tz)	hm(dawn_civil(38.8048, -77.0469, 2026, 7, 17, -4))	"05:26"
dawn_nautical / dusk_nautical(lat, lon, y, m, d, tz)	hm(dawn_nautical(38.8048, -77.0469, 2026, 7, 17, -4))	"04:48"
dawn_astro / dusk_astro(lat, lon, y, m, d, tz)	hm(dusk_astro(38.8048, -77.0469, 2026, 7, 17, -4))	"22:23"
solar_noon(lon, y, m, d, tz)	hm(solar_noon(-77.0469, 2026, 7, 17, -4))	"13:14"
day_length(lat, lon, y, m, d)	day_length(46.0569, 14.5058, 2026, 7, 17)	15.3443
sun_position(lat, lon, y, m, d, hour, tz)	sun_position(38.8048, -77.0469, 2026, 7, 17, 13.2, -4).alt	72.2914
moon_age(y, m, d)	moon_age(2026, 7, 17)	2.70686
moon_illum(y, m, d)	moon_illum(2026, 7, 17)	0.0806581
hm(h)	hm(6.05)	"06:03"
places	places.duluth_ga.lat	34.0029
drive_daylight(from, to, y, m, d, tz)	drive_daylight(places.alexandria_la, places.duluth_ga, 2026, 7, 17, -4).window_hours	15.8284

5. rmt.nu — random matrices, structured

packages/rmt.nu is sugar over randn/qr/eye for the matrices you actually reach for at the prompt: symmetric, positive definite (chol-safe by construction: eigenvalues at least 1), Haar orthogonal, permutations and their matrices, correlation, row-stochastic (random Markov chains), and the Gaussian orthogonal ensemble — whose spectrum follows Wigner's semicircle on $[-2, 2]$, as the last line demonstrates on a 200×200 draw. Everything is reproducible under `rng(seed)`.

```
neutrino> load("packages/rmt.nu")
neutrino> format(6)
neutrino> rng(11)
neutrino> randorth(2)
[0.684752, 0.728776; -0.728776, 0.684752]
neutrino> let P = randspd(3); chol(P)[1, 1] > 0
true
neutrino> randperm(6)
[4, 5, 1, 2, 3, 6]
neutrino> permuat([3, 1, 2])
```

```
[0, 0, 1; 1, 0, 0; 0, 1, 0]
neutrino> let H = goe(200); max(abs(eig(H).values))
1.99476
```

Function	Worked example	Result
randsym(n)	let S = randsym(3); max(max(abs(S - S')) == 0	true
randspd(n)	let P = randspd(3); min(eig(P).values) >= 1	true
wishart(n)	let W = wishart(3); min(eig(W).values) > 0	true
randorth(n)	let Q = randorth(3); max(max(abs(Q' * Q - eye(3)))) < 1e-12	true
randperm(n)	sort(randperm(5)) == 1:5	[true, true, true, true, true]
permmat(p)	permmat([2, 3, 1])	[0, 1, 0; 0, 0, 1; 1, 0, 0]
randcorr(n)	let C = randcorr(3); max(abs(diag(C) - 1)) < 1e-12	true
randstoch(n)	let T = randstoch(4); max(abs(sum(T, 2) - 1)) < 1e-12	true
goe(n)	let H = goe(150); max(abs(eig(H).values)) < 2.4	true

6. phys.nu — physical constants

packages/phys.nu is the CODATA 2018 constants as a record — exact where the 2019 SI redefinition makes them exact (c, h, k, NA, qe), the recommended values elsewhere. The transcript computes Earth's surface gravity from G, recovers the speed of light from Maxwell's relation $1/\sqrt{\epsilon_0\mu_0}$, and expresses thermal energy at room temperature in electron volts:

```
neutrino> load("packages/phys.nu")
neutrino> format(6)
neutrino> phys.c
299792458
neutrino> phys.hbar
1.05457e-34
neutrino> phys.G * 5.972e24 / 6.371e6 ^ 2
9.81997
neutrino> 1 / sqrt(phys.eps0 * phys.mu0)
2.99792e+08
neutrino> phys.k * 300 / phys.eV
0.0258520
```

Function	Worked example	Result
phys.c	phys.c == 299792458	true
phys.h	phys.h	6.62607e-34
phys.hbar	abs(phys.hbar * 2 * pi - phys.h) < 1e-45	true
phys.G	phys.G	6.67430e-11
phys.k (thermal, in eV)	phys.k * 300 / phys.eV	0.0258520

Function	Worked example	Result
phys.NA	phys.NA == 6.02214076e23	true
phys.R	phys.R	8.31446
phys.qe	phys.qe	1.60218e-19
phys.alpha	1 / phys.alpha	137.036
phys.sigma	phys.sigma	5.67037e-08
phys.ly / phys.au	phys.ly / phys.au	63241.1

7. scatter.nu — scatter plots without touching the core

Proof that the frozen language plots more than it appears to: `plot()`'s `style = "points"` option is honored by every backend (SVG circles in the browser, point markers in `ascii` and `gnuplot`), so scatter plots are a pure package. `scatter(x, y)` and `scatter_titled(x, y, t)` wrap that fact; `jitter(x, amount)` spreads overplotted values for legibility. Per-point sizes and colors would need backend changes and are deliberately absent — the package's header says so. The transcript exercises the numeric helper; the SVG output itself is asserted by the plot test suite:

```
neutrino> load("packages/scatter.nu")
neutrino> rng(1); let j = jitter(1:100, 0.1);
neutrino> max(abs(j - (1:100))) <= 0.05
true
neutrino> size(jitter(rand(3, 4), 0.2))
[3, 4]
```

Function	Worked example	Result
<code>jitter(x, a)</code> stays within <code>a/2</code>	<code>max(abs(jitter(1:100, 0.1) - (1:100))) <= 0.05</code>	true
<code>jitter</code> preserves shape	<code>size(jitter(rand(3, 4), 0.2)) == [3, 4]</code>	[true, true]
mean displacement is small	<code>abs(mean(jitter(zeros(1, 2000), 0.2))) < 0.01</code>	true

8. symb.nu — symbolic differentiation

A symbolic differentiator in pure Neutrino, addressed in mathematics and answering in it. The front door is a string:

```
neutrino> format(6)
neutrino> load("packages/symb.nu")
neutrino> deriv("sin(x)/x")
"((cos(x) / x) - (sin(x) / x^2))"
neutrino> deriv("x^x")
"(exp((x * log(x))) * (log(x) + (x / x)))"
neutrino> deriv("exp(-x^2)")
"(-2 * (exp((-x^2)) * x))"
neutrino> let f = dfun("sin(x)/x");
neutrino> fzero(f, 4, 5)
4.49341
neutrino> fminbnd(ffun("sin(x)/x"), 4, 5)
{x = 4.49341, fx = -0.217234}
neutrino> integral(dfun("x^3"), 1, 2)
7.00000
```


deriv parses, differentiates, simplifies, and prints; ffun and dfun lift a string to an ordinary function (of the original, or of its derivative), which then rides the whole numeric ecosystem — the transcript above finds the $\tan x = x$ critical point of $\sin(x)/x$ twice over (fzero on the symbolic derivative, fminbnd on the original, agreeing at 4.49341) and checks the Fundamental Theorem of Calculus (the integral of dfun("x^3") over [1, 2] is 7, the function's change).

How it works. Expressions are nested records; ddx differentiates by structural recursion. sub and divx desugar into add/mul/negative powers at construction, so the product, power, and chain rules alone carry the whole calculus — the quotient rule falls out of $d(b^{-1})$ for free, which is why deriv("sin(x)/x") above prints the textbook form. The parser is recursive descent over s[i] (character classes are chained comparisons: "0" <= c <= "9"), unary minus binds looser than ^ so -x^2 means $-(x^2)$, and general powers desugar as $f^g = \exp(g \cdot \log f)$ — which is how x^x differentiates with no special case. The printer recognizes division and subtraction; simp folds and hoists constants. Since v2.19.2 the dispatchers are written with elseif — the package whose nine-deep end-cascades helped motivate the feature was first to enjoy it.

The constructor layer remains public — it is the representation, and remains the door for programmatic tree-building: C(5), X, add, mul, powc, sinx, ... plus evalx (evaluate at a point), subst (compose), dn (n-th derivative), and taylor (series coefficients by repeated ddx). The transcript cross-checks the symbolic answer against numeric differencing — two independent derivatives agreeing inside one verified session — and recovers the sine series (0, 1, 0, $-1/6$, ...) from pure record recursion:

```
neutrino> format(4)
neutrino> load("packages/symb.nu")
neutrino> let e = add(powc(X, 3), mul(C(5), sinx(X)));
neutrino> show(simp(ddx(e)))
"((3 * x^2) + (5 * cos(x)))"
neutrino> let d = fn f -> fn x -> (f(x + h) - f(x - h)) / (2 * h) where h = 1e-6
<fn/1>
neutrino> abs(evalx(ddx(e), 2) - d(fn t -> t ^ 3 + 5 * sin(t))(2)) < 1e-8
true
neutrino> show(dn(powc(X, 5), 3))
"(60 * x^2)"
neutrino> taylor(sinx(X), 7)
[0.000, 1.000, -0.000, -0.1667, 0.000, 0.008333, -0.000, -0.0001984]
```

(History, kept honest: v2.12.0 shipped constructor-only under a recorded limitation — “no sub-string access” — that v2.13.1 discovered was false all along: strings index like arrays, golden-tested since the strings phase. v2.14.0 built the parser that was always possible; v2.15.0 taught the printer textbook manners; v2.16.0 added the function-space lifts. The wrong entry and its correction are both in KNOWN_LIMITATIONS — the goldens outrank the maintainer’s memory.)

Function	Worked example	Result
d/dx of x^3 at 2 is 12	evalx(ddx(powc(X, 3)), 2)	true
chain rule through exp(2x)	abs(evalx(ddx(subst(expx(X), true mul(C(2), X))), 0) - 2) < 1e-12	
taylor of exp begins 1, 1, 1/2	max(abs(taylor(expx(X), 2) - [1, 1, 0.5])) < 1e-9	true
parse then evaluate	abs(evalx(parse("2*pi"), 0) - 2 * pi) < 1e-12	true
deriv: string in, string out	deriv("x^2") == "(2 * x)"	true

Function	Worked example	Result
dfun: derivative as a function	<code>abs(dfun("x^3")(2) - 12) < 1e-12</code>	<code>true</code>
FTC: integral of dfun is the change	<code>abs(integral(dfun("x^3"), 1, 2) - 7) < 1e-6</code>	<code>true</code>

9. demo.nu — the tour

Not a library: a performance. `load("packages/demo.nu")` plays a guided tour in six acts — executable mathematics (Basel, the Gaussian integral), the blackboard's word order (where, chained masks), functions as values (anonymous application, Euclid's recursion, n-fold self-composition converging on the golden ratio), the pipelines act with both crown jewels (the oscillation pipe — values flavor-changing between `~>` and `|>` down one chain — and the fan-out dashboard), the calculus (symb.nu's `deriv` and a symbolic derivative fed to `fzero`), money (the HP-12C mortgage), and a plotted finale. Every number is computed live in the session, seeded where random, deterministic throughout; `make test` asserts the tour plays clean end to end. Point a newcomer at this file first.

```
neutrino> load("packages/demo.nu")
```

Writing your own

The pattern all four follow: a file of `let` definitions, `assert`-validated inputs, a record of closures when a namespace helps, and a golden test file whose reference values come from an independent implementation. Multi-line definitions are fine wherever a bracket is open. `save` your workspace, `load` it tomorrow — a package is just a workspace you chose to keep.